Huawei Access Node QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Resolution SOP

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4008

Description

This Standard Operating Procedure is designed to address critical QoS congestion events occurring on Huawei Access Nodes, specifically WHI-ACC-50 located at the Whitefield East POP. The failure domain encompasses microbursts, interface tail drops, and QoS queue starvation, all triggered by the ALARM_4008 code. The underlying protocol mechanics driving this issue are rooted in the Huawei Access Node's reliance on Hierarchical Quality of Service (H-QoS) architecture, which maps QoS policies onto a hierarchical structure of five classes. However, when the access node's QoS policies are misconfigured or exceed the capacity of the hardware, congestion events may occur, causing interface tail drops and microbursts. This can lead to QoS queue starvation, which in turn triggers the ALARM_4008 code, indicating a critical failure.

Pre-requisite Checks

- Verify the Whitefield East POP's physical infrastructure, including the access node's power supply, cooling system, and cabling.

- Check the redundant link status to ensure it is operational and not contributing to the congestion event.

- Verify the Huawei Access Node's firmware is up-to-date and matches the recommended version for WHI-ACC-50.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies exceeding hardware capacity	High
Hardware failure or degradation affecting QoS processing capabilities	Medium
Inadequate or malfunctioning QoS monitoring and policing mechanisms	Low

Diagnostic Commands & Expected Outputs

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Command: display qos policy, Expected Output: Display QoS policy configuration, including Hierarchical QoS (H-QoS) settings., Description: This command will reveal the QoS policies applied to the access node, allowing for an initial assessment of potential misconfigurations or over-subscription.

Command: display interface queue statistics, Expected Output: Display interface queue statistics, including queue depth, packet drops, and
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transmission errors., Description: This command provides insight into the interface's congestion levels, helping to identify areas where QoS queue starvation is occurring.

Command: display qos queue configuration, Expected Output: Display QoS queue configuration, including queue size, scheduling priority, and shaping rates., Description: This command will reveal the QoS queue settings, allowing for an assessment of whether the queues are properly configured to handle the expected traffic load.
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Resolution Steps

Step 1: Step: Update QoS Policies to Avoid Over-Subscription, Command: qos policy update, Description: Update the QoS policies to reflect the access node's true capacity and avoid over-subscription. This may involve re-configuring the Hierarchical QoS (H-QoS) settings to ensure they align with the access node's capabilities.

Step 2: Step: Adjust QoS Queue Settings for Optimal Performance, Command: qos queue update, Description: Adjust the QoS queue settings to optimize performance, ensuring queues are properly sized and configured for the expected traffic load. This may involve adjusting queue sizes, scheduling priorities, and shaping rates.

Step 3: Step: Verify QoS Monitoring and Policing Mechanisms, Command: display qos monitoring and policing, Description: Verify that QoS monitoring and policing mechanisms are adequately configured and functioning correctly. This may involve checking for any malfunctions or inadequacies in the QoS monitoring and policing processes.

Post-Check Verification

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Command: display qos policy, Expected Output: Verify updated QoS policy configuration, ensuring correct Hierarchical QoS (H-QoS) settings., Description: This command will confirm whether the updated QoS policies have been successfully applied and whether the Hierarchical QoS (H-QoS) settings align with the access node's capabilities.

Command: display interface queue statistics, Expected Output: Verify improved interface queue statistics, indicating reduced congestion levels and QoS queue starvation., Description: This command will provide insight into the interface's congestion levels, confirming whether the updated QoS policies and queue settings have successfully alleviated QoS queue starvation.

Command: display qos queue configuration, Expected Output: Verify updated QoS queue configuration, ensuring proper queue sizes, scheduling priorities, and shaping rates., Description: This command will confirm whether the updated QoS queue settings have been successfully applied and whether the queues are properly configured for optimal performance.
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Roll-back Procedures

- Step: Restore Original QoS Policies, Command: qos policy restore, Description: Restore the original QoS policies, reversing any changes made during the resolution process. This will ensure the access node reverts to its original configuration, preventing any potential disruptions or misconfigurations.

- Step: Revert QoS Queue Settings, Command: qos queue restore, Description: Revert the QoS queue settings to their original configuration, ensuring the queues return to their pre-resolution state. This will prevent any potential disruptions or misconfigurations.